



CONERGY

String inverters | Technical data

Conergy IPG 2000-5000 (vision)

The newly developed string inverters of the Conergy IPG series set a new standard for single-phase, grid-connected inverters. As the result of many years of experience and technical expertise of Conergy in developing and manufacturing solar inverters, they meet the highest demands for performance, functionality and design.

Conergy IPG string inverters

The string inverters of the new Conergy IPG series, which are available in the 2000, 3000, 4000 or 5000 watt performance class, are an ideal match for small to medium power photovoltaic systems.

Conergy IPG string inverter vision with touch-screen display

The Conergy IPG string inverter vision features a standard high-resolution touch-screen display. All important information on the inverter and the PV generator can be accessed on the display by touch. This gives the user the ability to see the exact operating status of the system at all times. Up to 6 IPG inverters can be read out via one display.

Maximum yields

With a maximum efficiency rate of 96.7 % and a European-standard efficiency rate of 96 %, the string inverters of the Conergy IPG series can supply AC current to the public power grid in a highly efficient manner thanks to advanced transformerless technology. An accurate and fast MPP tracking function also provides additional power optimisation.

Balanced Mode Technology – Greater security and longer service life

The Balanced Mode Technology (BMT) circuit concept patented by Conergy reduces leakage currents to below the limit values prescribed for home installation. Thanks to this switching-principle, the transformerless inverter guarantees a gentle operation with all kinds of PV modules including sensitive thin-film modules. Therefore, the general PV plant safety is increased as a possible disconnection of the ground fault circuit interrupter (GFCI) cannot arise. This unique feature is achieved without having to make a compromise concerning the inverter's efficiency, which is maximised by transformerless conversion.



Freedom in designing the solar generator

The extremely wide solar generator input voltage range of all Conergy IPG string inverters ($U_{DC} = 220-800 \text{ V}$) provides almost limitless wire connection possibilities with PV modules. The high input voltage allows maximum string length for cost-effective and quick wire connection to PV modules.

Fast assembly and installation

The Conergy IPG string inverter can be quickly mounted to a wall by only one person using the enclosed wall bracket. The pre-assembled plugs and connection lines let you easily connect the solar generator via two MC4 plugs. Two strings can be connected in this way by default. If necessary, additional strings can be connected using a Y-branch.

At home in any environment

With an IP 65 protection type and an ambient temperature range of -10 to $+60 \text{ }^{\circ}\text{C}$, installation can be carried out in almost any environment. The Conergy IPG string inverter can also be installed in the living area of any home thanks to its aesthetic design and silent operation (fanless cooling).



PHOTOVOLTAICS

**Conergy IPG 2000-5000 (vision)****CONERGY****Accessories: Conergy EasyConnect**

The combination of a single-phase inverter from the Conergy IPG series and the optional Conergy EasyConnect connection box significantly eases the planning and installation of the entire solar power system. The Conergy EasyConnect connection box bundles external PV system components into one device: an electronic DC disconnect, a category B and C lightning protection and the Conergy SunReader system monitoring device. Since all components and plug connections are pre-assembled, both devices can be connected to each other easily and trouble-free.



	Conergy IPG 2000	Conergy IPG 3000	Conergy IPG 4000	Conergy IPG 5000
MPP voltage range	200–680 V	200–680 V	220–680 V	220–680 V
Max. DC voltage (U_{DCmax})	800 V _{DC}	800 V _{DC}	800 V _{DC}	800 V _{DC}
Max. input current (I_{DCmax})	10.2 A	10.2 A	15.2 A	16.2 A
Mains voltage (U_{AC})	196–253 V _{AC}	196–253 V _{AC}	196–253 V _{AC}	196–253 V _{AC}
Frequency (f_{AC})	47.5–50.2 Hz	47.5–50.2 Hz	47.5–50.2 Hz	47.5–50.2 Hz
Nom. AC power (P_{ACnom})	1,800 W	2,600 W	3,400 W	4,600 W
Max. AC power (P_{ACmax})	2,000 W	2,800 W	3,800 W	5,000 W
Distortion factor	< 3 %	< 3 %	< 3 %	< 3 %
Max. efficiency	96.0 %	96.0 %	96.5 %	96.7 %
European standard efficiency	93.6 %	94.5 %	95 %	96 %
Physical dimensions (W x H x D)	377 x 620 x 226 mm	377 x 620 x 226 mm	377 x 620 x 226 mm	377 x 620 x 226 mm
Weight	23 kg	24 kg	27 kg	28 kg
Cooling	Convection	Convection	Convection	Convection
Ambient temperature	–10 to +40 °C/up to +60 °C with derating			
Protection type	IP 65	IP 65	IP 65	IP 65
Earth fault monitoring	Yes	Yes	Yes	Yes
DC overvoltage protection	Varistors and spark gaps			
Reverse polarity protection	DC-side short-circuit diodes			
DC overload behaviour	Shifting of the working point on the solar generator characteristic curve			
LED display	Yes	Yes	Yes	Yes
Touch-screen display	Only with Conergy IPG vision ¹			
Max. number of strings	2	2	2	2
DC connect	MC IV	MC IV	MC IV	MC IV
Communication	CAN-Bus	CAN-Bus	CAN-Bus	CAN-Bus

¹ A lot of plant data can be called up via the display. This may slightly affect the efficiency.

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